

PROJECT EXPO-2026

SMART WILDPIG DETECTION AND ALERT SYSTEM USING IoT AND AI

TEAM MEMBERS:

DHANYA SRI J

LAVANYA V

ARUNACHALAM M

SUJITH N

Phase5-Testing and Validation Review (TVR)

Functional Testing Report:

Test ID	Test Description	Expected Result	Actual Result	Status
FT-01	PIR detects movement	Motion signal is generated.	Motion signal is generated.	Pass
FT-02	ESP8266 receives PIR trigger	ESP8266 responds to the trigger.	ESP8266 responds to the trigger.	Pass
FT-03	ESP32-CAM receives trigger	Camera starts the capture process.	Camera starts the capture process.	Pass
FT-04	Image capture	Clear image is captured.	Clear image is captured.	Pass
FT-05	Image transfer to PC	Captured image reaches the PC.	Captured image reaches the PC.	Pass
FT-06	MobileNetV2 classification	Image is classified correctly.	Image is classified correctly.	Pass
FT-07	Wild pig detected	LED and buzzer turn ON.	LED and buzzer turn ON.	Pass
FT-08	Non-pig image	No wild-pig alert is generated.	No wild-pig alert is generated	Pass
FT-09	Pig leaves / no longer detected	LED and buzzer turn OFF.	LED and buzzer turn OFF.	Pass
FT-10	Resume monitoring	System returns to monitoring mode.	System returns to monitoring mode.	Pass

Prototype Under Test:

PIR Sensor:

- Detects movement in the agricultural field and provides a trigger.

ESP8266 NodeMCU:

- Receives the PIR trigger and coordinates communication/control.

ESP32-CAM:

- Captures the image after motion is detected.

PC / Antigravity:

- Receives/processes the captured image and runs the computer-vision workflow.

MobileNetV2:

- Classifies the captured image to identify whether a wild pig is present.

LED:

- Provides a visual alert when a wild pig is detected.

Buzzer:

- Provides an audible alert when a wild pig is detected.

Arduino IDE:

- Used for embedded programming and uploading/testing the microcontroller code.

Performance Evaluation:

Metric	Result
Training Accuracy	99.59%
Validation Accuracy	98.94%
Test Accuracy	99.05%

Dataset Distribution:

Dataset	Pig	Non-Pig	Total
Training	653	826	1479
Validation	117	168	285
Testing	233	84	317
Total	1003	1078	2081

Testing Case:

TC ID	Test Condition	Expected Result	Result
TC-01	Wild pig in daylight	Wild pig should be identified and alert activated.	Wild pig should be identified and alert activated.
TC-02	Wild pig under low-light conditions	System should attempt image capture and classification.	System should attempt image capture and classification.
TC-03	Human movement	System should avoid generating a wild-pig alert.	System should avoid generating a wild-pig alert.
TC-04	Dog/other animal movement	System should avoid generating a wild-pig alert.	System should avoid generating a wild-pig alert.

TC-05	No movement	System remains in monitoring mode.	System remains in monitoring mode.
TC-06	Wild pig at different distance	Detection performance should be evaluated.	Detection performance should be evaluated.
TC-07	Partially visible pig	Classification performance should be evaluated.	Classification performance should be evaluated.
TC-08	Repeated motion events	System should respond repeatedly without failure.	System should respond repeatedly without failure.
TC-09	Communication interruption	Failure should be identified and system recovery evaluated.	Failure should be identified and system recovery evaluated.
TC-10	Continuous operation	System should operate reliably for the selected test duration.	System should operate reliably for the selected test duration.

Error Analysis:

Error / Limitation	Possible Cause	Effect	Proposed Action
False PIR trigger	Human, dog, or other movement	Unnecessary image capture	Improve AI verification and sensor placement.
Poor image quality	Low light, rain, fog, or distance	Reduced classification accuracy	Improve camera placement and lighting.
False AI detection	Similar-looking objects/animals	Incorrect alert	Increase and improve the training dataset.

Missed detection	Pig partly hidden or outside camera view	No alert	Optimize camera position and detection conditions.
Communication failure	Connection or serial/network issue	Image not processed	Check connections and add recovery handling.
Power instability	Insufficient or unstable supply	Reset or malfunction	Use a stable regulated power source.

Debugging Report:

Problem Observed	Likely Cause	Debugging / Corrective Action	Status
PIR not detecting movement	Incorrect VCC/GND/OUT connection	Verify wiring and PIR output.	Verified wiring and PIR sensor
ESP8266 not responding	Code, power, or serial issue	Check upload, serial monitor, and power.	Checked the uploads, serial monitor, and power
ESP32-CAM not capturing	Camera configuration/power issue	Check camera configuration, connections, and supply.	Checked camera configuration, connections, and supply.
LED not glowing	Incorrect GPIO/wiring	Verify GPIO, polarity, and resistor.	Verified GPIO, polarity, and resistor.
Buzzer not sounding	Incorrect wiring/code	Verify buzzer connection and control logic.	Verified buzzer connection and control logic.
Image not processed	Transfer/software issue	Check PC connection and processing workflow.	Checked the PC connection and processing workflow.
Incorrect MobileNetV2 result	Dataset/model limitation	Improve dataset and evaluate model.	Improved dataset and evaluate model.

Future Improvements

- Increase the size and diversity of the wild-pig image dataset.
- Improve model accuracy and reduce false positive and false negative detections.
- Improve night-time and adverse-weather image capture.
- Add IoT/mobile notification for remote farmer alerts.
- Use a weather-resistant enclosure for field deployment.
- Use solar power or battery backup for outdoor operation.
- Move AI processing to a suitable edge-AI platform in a future version to reduce dependence on a PC.
- Design a custom PCB and improved mechanical mounting for the final prototype.

